

JATIN

SNOWFLAKE DATA ENGINEER, 7+

Summary:

Snowflake Data Engineer with 7+ years of experience designing, building, and optimising production-grade cloud data warehouses, ELT pipelines, and data models across procurement, financial services, IP analytics, and healthcare domains. Deep practical experience with Snowflake (Streams, Tasks, Zero-Copy Cloning, Clustering Keys, Dynamic Data Masking, RBAC, Data Sharing), DBT (staging/intermediate/mart layers, incremental models, tests, macros), and Apache Airflow orchestration. Strong Python engineer with proven delivery across Azure and AWS cloud environments.

SKILLS

Snowflake: Streams & Tasks, Zero-Copy Cloning, Clustering Keys, Materialized Views, Result Caching, Dynamic Data Masking, Row Access Policies, RBAC, Data Sharing, Snowpipe, Query
DBT: Optimization, Warehouse Cost Management, Auto-Suspend

Data Modelling: version-controlled SQL — dbt Core & dbt Cloud Star Schema, Snowflake Schema, Data Vault, SCD Type 2, fact & dimension tables, OBT, normalised & denormalised designs

ELT / ETL: Python-based ingestion pipelines, REST API extraction, schema mapping, currency conversions, fiscal year alignment, semi-structured data (JSON, XML, VARIANT, FLATTEN) Apache Airflow

EXPERIENCE

CONFIDENTIAL

As Consultant — Data Engineer

- Procurement Data Platform · Snowflake + DBT + Airflow · ERP Integration
- Architected end-to-end procurement data warehouse on Snowflake: designed fact tables for Purchase Orders, GRNs & Invoices with dimension tables for vendors, cost centres, categories & delivery locations — enabling real-time visibility into vendor performance and supply chain analytics.
- Built full DBT transformation layer: staging models for raw ERP ingestion, intermediate models for business logic, and mart models for spend analytics, vendor scorecards & PO cycle time reporting — all version-controlled in Git with dbt tests on every model.
- Integrated procurement data from SAP/Oracle ERP systems into Snowflake: Python-based ingestion pipelines handling complex nested JSON structures, schema variations, and incremental loads via Snowflake Streams & Tasks.
- Developed vendor performance scorecards: tracking on-time delivery rates, quality rejection rates & payment terms compliance across 100+ vendors — surfaced via Snowflake-backed reporting layer.
- Implemented Role-Based Access Control in Snowflake: procurement managers, finance teams & executives each see only their relevant data — enforced via row access policies and dynamic data masking.
- Optimised Snowflake compute costs: warehouse auto-suspend policies, query profiling, clustering key tuning & result caching — reducing monthly Snowflake spend by 35%.

(DAGs, retry logic, SLA monitoring, Slack alerting), Azure Data Factory, dbt job scheduling

Orchestration: Apache Airflow (DAGs, retry logic, SLA monitoring, Slack alerting), Azure Data Factory, dbt job scheduling

Data Quality: Automated reconciliation frameworks, dbt tests, Great Expectations, audit trail frameworks, zero-tolerance validation scripts

Cloud & Storage: Azure (Databricks, ADLS Gen2, ADF, Key Vault) · AWS (S3, Glue, EC2) · Snowflake on Azure/AWS

Python & SQL: Python 3, Pandas, PySpark, Snowflake SQL, stored procedures, views, window functions, CTEs, DuckDB, SQLAlchemy

Governance & Security: Dynamic Data Masking, Row Access Policies, RBAC role hierarchies, PHI masking, HIPAA compliance, audit logging, Unity Catalog

Databases: Snowflake, Azure Databricks (Delta Lake), PostgreSQL, MySQL, MongoDB, SQLite, SQL Server

Version Control: Git, GitHub, branching strategies, dbt project version control

- Orchestrated all pipelines via Apache Airflow: DAGs for daily ERP ingestion, dbt runs & data quality checks with Slack alerting on failures and full retry logic.

PROJECTS

1. Procurement Data Warehouse — Snowflake + DBT + Airflow (StatusNeo)

Stack: Snowflake · DBT · Apache Airflow · Python · SAP/Oracle ERP APIs · Git · Slack

- End-to-end procurement DWH: fact tables (POs, GRNs, Invoices) + dimension tables (vendors, cost centres, categories); full DBT model layer with staging → intermediate → mart, dbt tests on every model, Git version control.
- 100+ vendor scorecards: on-time delivery, rejection rates, payment terms compliance — live in Snowflake-backed reporting layer; 35% Snowflake cost reduction via auto-suspend + query optimisation.
- Airflow DAG orchestration: daily ERP ingestion + dbt runs + data quality checks; Slack alerting on failures; Snowflake RBAC ensuring role-based data visibility for procurement, finance & execs.

2. Financial Data Migration — QuickBooks → Snowflake (ISSC)

Stack: Snowflake · Python · QuickBooks API · Apache Airflow · SCD Type 2 · SQL

- Zero data loss migration of GL, AP/AR, invoices, payroll & trial balance data into Snowflake financial DWH: SCD Type 2 for full audit trail; automated Python reconciliation framework achieving 100% balance accuracy.
- CFO-ready P&L, balance sheet & cash flow reporting layer: directly queryable in Snowflake; Airflow-orchestrated with full error handling, retry logic, and execution logging.

3. Global Patent Analytics Platform — Snowflake + DBT (Clarivate)

Stack: Snowflake · DBT · Apache Airflow · AWS S3 · Python · SQL

- Unified 50+ global PTO offices into single Snowflake DWH: Star Schema for inventors/assignees/IPC classifications; Snowflake Streams & Tasks for incremental ELT (72 hrs → 4 hrs freshness).
- 65% query performance improvement: via Clustering Keys, Materialized Views & Result Caching on billion-row datasets; ZeroCopy Cloning for dev/test; Row Access Policies + Dynamic Data Masking for tiered client licensing.
- DBT transformation layer: modular staging/intermediate/mart models, dbt tests, version-controlled SQL; Airflow orchestration with 99.8% uptime SLA.

EDUCATION

B.Tech — Computer Science & Engineering | GL Bajaj Group of Institutions, Mathura | Aug 2018 – Jul 2022

4. Global Clinical Data Platform — Snowflake + Databricks + FHIR (Agilon Health)

Stack: Snowflake · Azure Databricks · PySpark · Apache NiFi · Python · AWS S3 · Azure ADLS · FHIR R4 · HL7

- Consolidated EHR/EMR data from Epic, Cerner & Meditech across US, Australia & Brazil into Snowflake clinical DWH: FHIR R4 pipelines, PHI masking, field-level encryption, HIPAA/LGPD compliance.
- PySpark Medallion pipelines (Bronze/Silver/Gold): millions of patient records; automated Python data quality framework saving 20+ hrs/week; patient risk scoring & population health dashboards on Snowflake datamart.